

A Functional Overview of Integration

UChicago Political Science Math Camp

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2025

Some Thoughts On Integration

- ▶ Integration is the inverse of derivation, but can be conceptually more challenging
- ▶ Integral rules are often more difficult than derivative rules
- ▶ This session is more concerned with how to understand integrals than with how to compute them (i.e., intuition over execution)
- ▶ “Spaghetti at the wall” pedagogical approach

Expectation

When calculating expected outcomes, we employ the following formula:

$$E[u] = \sum_i \text{Pr}[i] \cdot u_i$$

The *utility* of an event, then, is its value times its probability.

Let *Heads* = 1 and *Tails* = 0. What is the expected value of a coin toss?

More Expectation

We could model the expected roll of a D4 as

$$\sum_i \Pr[i] \cdot u_i = \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{4} \cdot 3 + \frac{1}{4} \cdot 4 = 2.5$$

What about a D6?

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A D10?

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A D100?

Something Weird About Probability

What if we keep finding larger (fair) dice? What is $\lim_{i \rightarrow \infty} \Pr[x = i]$?

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What if we keep finding larger (fair) dice? What is $\lim_{i \rightarrow \infty} \Pr[x = i]$?

Assume that every outcome is equally likely (uniform distribution). We can evaluate:

$$\sum_i^{\infty} \Pr[i] \cdot u_i = \sum_i^{\infty} \frac{1}{\infty} u_i$$

And our summation and series rules lend that

$$\Rightarrow \frac{1}{\infty} \cdot \sum_i^{\infty} u^i \Rightarrow \text{undefined}$$

Something's not right here...

A Qualifier

Let's take away the dice's fairness add a new condition:

Every face of the dice has a unique value in $[-b, b]$

Then, as $n_{\text{face}} \rightarrow \infty$, every $x \in [-b, b]$ has a corresponding dice roll.

Calculus and Infinity

Calculus fundamentally solves the following contradictions:

- ▶ Derivation solved for the rate of change of a function as $\Delta x \rightarrow 0$, such that $\frac{\text{rise}}{\text{run}} \rightarrow \infty$.
- ▶ Integration solves for the sum of an infinite number of infinitely small segments, where the size of each approaches 0 such that $\sum_i \rightarrow 0$.

In most cases, $\frac{\text{rise}}{\text{run}} \neq \infty$ and $\sum_i \neq 0$.

A Different Approach to Intuition

Say we want to measure the area bounded by some function(s), or the **area beneath the curve**:

1. If all functions are linear, we can use geometric formulas (extensions of $A = bh$) to evaluate
2. If even one function is non-linear¹, then standard geometry can only approximate

¹At least, non-circular, although the area of a circle is derived from calculus anyways

Integration From Approximation

We can approximate the area under a curve by splitting the area into a series of rectangles, such that $A = \sum_i bh_i$

With very large rectangles (i.e., very large b -values), this creates a lot of error. **But what happens** as we add more rectangles, and shrink the size of each?

$$A = \sum_i bh_i \quad \implies \quad b \rightarrow 0, i \rightarrow \infty$$

How does this reflect our derivation definition?

Notation

Integrate $f(x)$ with respect to $x \rightarrow$ What is the sum of every $f(x)$ over every possible x ?

We denote this as

$$\int f(x) dx$$

We call this an **indefinite integral**

A New Limit Definition

$$\int f(x)dx = \lim_{b \rightarrow 0} \sum_i^{\frac{1}{b}} bf(x_i)$$

Integration vs Derivation

- ▶ The integral of $f(x)dx$ represents the sum of all f 's, such that if f represents a rate of change, then $\int f(x)dx$ represents the **change that occurred...**
- ▶ Think: what function's derivative returns $f(x)$? That function represents $\int f(x)dx$

To borrow again from Physics:

$$\text{Jerk} \Rightarrow \int dt = \text{Acceleration} \Rightarrow \int dt = \text{Speed} \Rightarrow \int dt = \text{Position}$$

Definite vs. Indefinite Integrals

Often, we only want to evaluate the integral of a function over some specific domain. To integrate $f(x)$ w.r.t x over the domain $[a, b]$, we write the **definite integral**

$$\int_a^b f(x)dx$$

An indefinite integral often produces a function. A definite integral almost always returns a number (or function independent of x).

$$\int_a^b f'(x)dx = f(b) - f(a)$$

This yields two truths:

1. The integral is the inverse of the derivative (the integral of a function's derivative is just the function)
2. For some defined domain $[a, b]$, the value of the integral over that domain is $\int f(b)dx - \int f(a)dx$

FTOC and Indefinite Integrals

Let's reiterate an important point here:

If we denote the indefinite integral $\int f(x)dx = F$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Then, to solve $\int_a^b f(x)dx$, we evaluate:

$$\int f(x)dx = F(x) \quad \implies \quad \int_a^b f(x)dx = F(x) \Big|_{x=a}^{x=b}$$

How Do We Actually Do This?



The Good Stuff

We can use a series of rules to integrate a function in its general form
for any value of x :

$$f(x) \implies [\text{RULE}] \implies \int f(x) dx$$

Note that now, our indefinite integral is itself another function of x .

The Sum Rule

For all integrals:

$$\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$$

For definite integrals such that $a \leq b \leq c$:

$$\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$$

The Power Rule, Backwards

Recall the derivative power rule:

$$\frac{d}{dx}x^k = k \cdot x^{k-1}$$

Then, if $\int f'(x)dx = f(x)$, what function would, when derived, produce ours?

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$$\int x^k dx = \frac{x^{k+1}}{k+1}$$

plus one little thing...

Remember Our Constants?

1. We know that the derivative of a constant is 0, such that $\frac{d}{dx}c = 0 \forall c \in \mathbb{R}$
2. If integration “reverts” a derivative, then it must account for any constant terms zero-d out in derivation
3. We resolve this as follows:

$$\int x^k dx = \frac{x^{k+1}}{k+1} + c$$

even though we don't know what c actually is.

A Power Rule Corollary (Reversed)

1. We know that $\int x^k dx = \frac{1}{k+1} x^{k+1}$ (**Power Rule**) and that $\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$ (**Sum Rule**)
2. We also know that for a constant ℓ , $\ell \cdot n = \sum_1^\ell n$, or n added ℓ times.
3. Then, the integral is the sum of x^k added up ℓ times:

$$\int \left[\ell \cdot x^k \right] dx = \sum_1^\ell \int x^k dx$$

4. We know that $\sum_1^n x = nx$, so

$$\sum_1^\ell \int x^k dx = \ell \cdot \int x^k dx \quad \Rightarrow \quad \int \left[\ell \cdot x^k \right] dx = \ell \cdot \frac{1}{k+1} \cdot x^{k+1}$$

$\Rightarrow$ **Coefficients:** The coefficient of a term is kept as a coefficient of that term's integral (i.e., multiply $\frac{1}{k+1}$ by the coefficient as well)

Quick Note on Constants

Recall that a constant m can be represented as $m \cdot x^0$. Then,

$$\begin{aligned}\int m dx &= \int m \cdot x^0 dx \\ \implies m \cdot \int x^0 dx \\ \implies m \cdot \frac{x^{0+1}}{0+1} + c &= m \cdot \frac{x^1}{1} + c \\ \therefore \int m dx &= mx + c\end{aligned}$$

Some Examples

Let's try some simple examples:

$$\int x^2 + 2x^3 dx$$

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$$\int 3 + \sqrt[3]{x} dx$$

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$$\int 2x^3 + \frac{1}{2}x^4 dx$$

$$\int 3 + \sqrt[3]{x} dx$$

$$\int x^{-3} + x^{-2} + x^{-1} dx$$

Closing Some Subtle Foreshadowing

The power rule doesn't work for $f(x) = x^{-1} = \frac{1}{x}$, as it solves:

$$\int x^{-1} dx = \frac{1}{-1+1} x^{-1+1} = -\frac{x^0}{0}$$

Thankfully, our natural logarithm solves this for us:

$$\int \frac{1}{x} dx = \ln(x) + c$$

Integrating Exponentials

We have a specific rule for integrating exponentials and logarithms:

$$\int a^x dx = \frac{a^x}{\ln a} + c$$

$$\int \log_b x dx = \frac{x}{\ln b} (\ln b - 1) + c$$

Definite Integration

So far, we've expressed $F(x)$ for every $f(x)$, both as functions of x , as *indefinite integrals*. When we add bounds of integration, we simply use FTOC:

$$\int_a^b f(x) dx = F(b) - F(a)$$

$$\int_2^4 x^3 dx = \left. \int x^3 dx \right|_2^4$$

$$\Rightarrow \left. \frac{x^4}{4} + c \right|_2^4$$

But, we don't know c ...

c you later

There's a handy trick with definite integrals:

$$\begin{aligned}\int f(x)dx &= F(x) + c \\ \implies \int_a^b f(x)dx &= F(b) + c - (F(a) + c) \\ \implies \int_a^b f(x)dx &= F(b) - F(a)\end{aligned}$$

For definite integrals, we subtract $c - c$ and can therefore exclude it. So,

$$\left. \frac{x^4}{4} + c \right|_2^4 = \frac{4^4}{4} - \frac{2^4}{4} = 16$$

More Examples

$$\int_0^2 x^3 + 2x^3 dx$$

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$$\int_2^6 \frac{1}{2}x^2 + 4 dx$$

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$$\int_0^b \sqrt{x} + \sqrt{2} dx$$

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$$\int_2^6 \frac{1}{2}x^2 + 4 dx$$

$$\int_0^b \sqrt{x} + \sqrt{2} dx$$

$$\int_a^2 \frac{1}{x^2} dx$$

More Examples

$$\int_0^2 x^3 + 2x^3 dx$$

$$\int_2^6 \frac{1}{2}x^2 + 4dx$$

$$\int_0^b \sqrt{x} + \sqrt{2}dx$$

$$\int_a^2 \frac{1}{x^2} dx$$

A Turn For The Worst

Two of our fundamental derivative rules – the **chain rule** and the **product rule** – are a bit more complicated to invert:

$$\frac{d}{dx}f(u) = f'(u) \cdot u'$$

$$\implies F(u) = \int f(u) \cdot u du \dots$$

There's not a clear solution here

U-Substitution

The inverse of the Chain Rule is called **integration by substitution**, or “**u-sub**”, and works as follows:

1. Express the integral in the form

$$\int f(g(x))g'(x)dx$$

2. Assign $g(x) = u$ such that the integral can be rewritten:

$$\int f(u)du$$

3. Evaluate the integral:

$$\int f(u)du = F(u)$$

4. Substitute back $u = g(x)$:

$$\int f(g(x))g'(x)dx = F(g(x))$$

Integration By Parts

Integration by parts inverts the product rule. We will not cover it in math camp, but you should be aware of the concept and ready to learn its process.

Area of a Circle

We know that $a = \pi r^2$, but the formal definition of π is only $\pi = \frac{c}{d}$.

Using *only* this definition for π , how might we calculate the area of a circle?

The Math

A circle can be divided into “slices,” and each slice can be approximated by a sum of triangles: $a_{\text{slice}} = \sum bh$. The error of these triangles approaches zero as we add more and more triangles...

Imagine we take the sum of the area of an infinite number of triangles, rotated around the center of a circle. Then, since our triangle area is $a = \frac{1}{2}bh$, we have

$$a_{\text{circle}} = \sum_i^{\infty} \frac{1}{i} \cdot \frac{1}{2}h$$

The “height” in this case is the radius of the circle. Now we have:

$$\sum_i^{\infty} \frac{1}{i} \frac{1}{2}r = \int_{\text{circle}} \frac{1}{2}r di = \frac{1}{2}r \cdot \int_{\text{circle}} di$$

We know that $\int di = i$, so we just need to choose our bounds. We are integrating over the total distance of the circumference, which we know to be $2r \cdot \pi$. So, we can place the bounds:

$$\int_0^{2r \cdot \pi} \frac{1}{2}r di = \frac{1}{2}ri \Big|_0^{2r \cdot \pi} = \frac{1}{2}r(2\pi r) - 0 = \pi r^2$$

Improper Integrals

For definite integrals which do not converge to $f(x) = a$, or for any asymmetric integrals whereby $f(x) = \pm\infty$ for $x \in \mathbb{R}$, we call that integral **improper**.

This can either result in a finite value, such as $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$, or an undefined value, such as $\int_{-\infty}^{\infty} \frac{1}{x} dx$

Gabriel's Horn

Imagine a horn formed by the function $f = \frac{1}{x}$ for the domain $x \in [1, \infty)$

What is its volume? What is its surface area?

Horn Volume

We know that $V = a \cdot h$; in this case we can set $h = \infty$ and $a = \pi r(x)^2 = \frac{\pi}{x^2}$. The volume is then the sum of this area over every possible height:

$$V = \int_1^{\infty} \frac{\pi}{x^2} dx = -\frac{\pi}{x} \Big|_1^{\infty}$$

$$V = \pi$$

The volume of the horn is finite.

Horn Surface Area

We know that surface area can be calculated as $s = c \cdot h$. In this case, we set $h = \infty$ and $c = \pi 2r = \frac{2\pi}{x}$. The surface area is then the sum of this circumference over every possible diameter:

$$s = \int_1^{\infty} \frac{2\pi}{x} dx = 2\pi \cdot \ln(x) \Big|_1^{\infty}$$

$$s = \infty$$

The horn has finite volume but infinite surface area. *You could fill the inside with paint, but never fully paint the outside.*